

Determining Probabilistic Load Flow Input Parameters Using Clustering Techniques

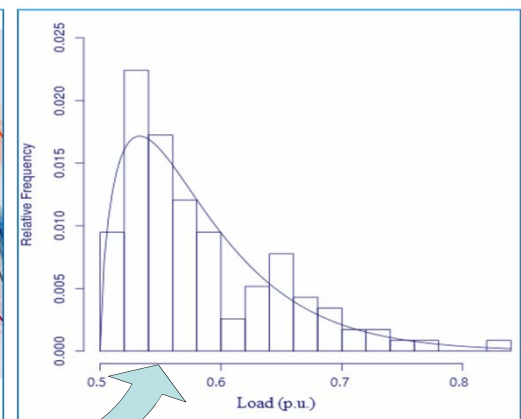
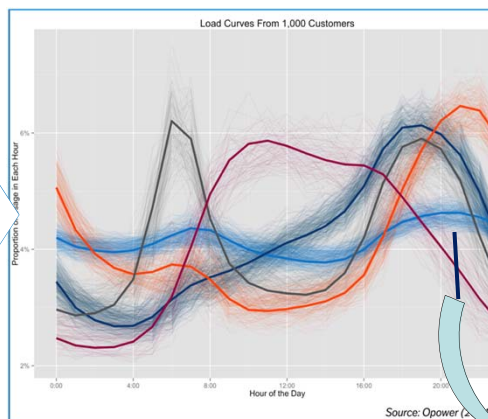
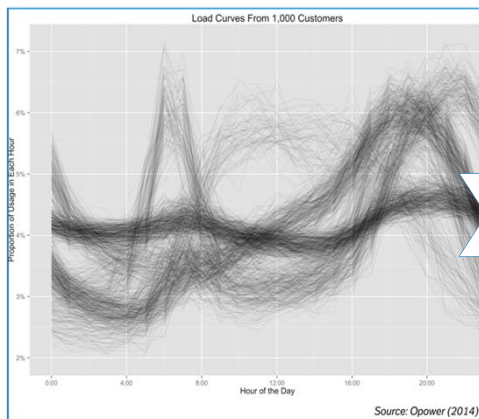
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Background

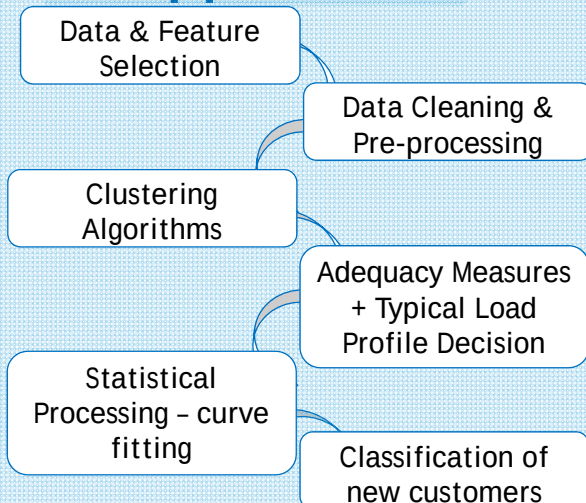
- Existence of uncertainty sources in power system – intermittent power generation, general stochastic customer load variations.
- Probabilistic Load Flow (PLF) approaches = more representative + better defined load flow results.
- PLF requires the description of customer loads/DG through pdfs or cdfs to calculate the range of probable system output states.
- Statistical load and generator profiles are required.

Clustering

- Many customers on the network – load model for each customer computationally impossible .
- 'Clusters' or customer groups with a Typical Load Profile (TLP) are needed.
- Clustering uses pattern recognition or analysis of proximity among load datasets to select groups.
- Methods have been used in power system load research to determine TLPs but rarely to derive statistical load parameters.



Approach



Conclusions

- Possibilities for generating appropriate statistical load models for PLF applications using clustering.
- Partitioning by seasons, day type (weekday / weekend) and specified hourly periods could allow a more meaningful clustering.
- Approach opens up practical usefulness of probabilistic approaches in power system analysis